Charotar University of Science and Technology

M.Sc (Advance Organic Chemistry) Semester- I Examination April 2011

OC -704: Analytical Techniques –I

Date: April 28, 2011 **Time: 10.00 am to 10.30 am**

Max. Marks:20

Part A

1. Explain the terms: E_R' and K (2)

2. What is meant by gradient elution in HPLC? (2)

3. What important properties a stationary phase must possess to be employed in GC? (2)

4. Explain the abbreviations NPC and RPC used in chromatography. (2)

5. Give the names of detectors used in HPLC. (2)

6. How does GLC differ from GSC? (2)

7. Mention few applications of GLC. (2)

8. What are the intensities of M and M+1 in the mass spectrum of methane? (2)

9. Explain the formation of m/z 43 in the mass spectrum of 2 methyl propane. (2)

10. What different isotopic combinations are possible for ethane ,taking into consideration the occurrence of C^{12} and C^{13} ? (2)

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Part B

1.(a). Explain Van Deemter equation. (3)

OR

(a).Draw the graph from the results given below and find out the optimum carrier gas velocity.

HETP(cm)	0.19	0.13	0.14	0.19
V(cm/sec)	0.71	1.38	2.70	6.12

(b). Draw a typical gas chromatogram and explain it. (3)

(c).The chromatographic peak for a compound A is found at 15.0 min(t_{RA})after sample injection and t_m is 2.0 sec. The peak width at the base is 24.2 sec for A. The column length is 40.2 cm. (4)

i.Calculate N.

ii From the method of preparation of column, it is found that V_s =9.9 ml and V_m=12.3 ml. Calculate the partition coefficient for A.

iii. What would be the resolution of A and compound B ,if the separation factor for B and A is 1.011?

2(a).Draw the block diagrams for GC and HPLC. (5)

(b).Write in brief on 'Solute Property Detector' used in LC (5)

OR

(b).How is GC employed to characterize hydrogen peroxide?

3(a).Discuss the head space method to determine halogenated hydrocarbons in H_2O. (3)

(b). Explain in brief , how combustion GC may be employed to determine the carbon content in H_2O_2 . (4)

(c).Explain the separation mechanism in LC. (3)

4.Answer any two from the following: (10)

a. Discuss the advantages and disadvantages of GLC and HPLC.

b.Write a note on electron capture detector (ECD).

c. Provide a brief account of the method to determine heat of vaporization of aromatic hydrocarbons.

5(a). Draw the molecular ions which may be formed from methyl vinyl ether. (2)

(b).Explain, how mass spectral data can distinguish between 1-propanol and 2-propanol. (4)

(c).Draw the ethyl bromide molecular ion. Explain the formation of ion at m/z 29 in the mass spectrum of ethyl bromide. (2)

(d).Give a schematic diagram of a mass spectrometer and explain the different operations. (2)